

Journal during their train ride to reflect properly on the attributes of random events. Or perhaps they watched too many gurus on television. Or perhaps they spent too much time upgrading their PalmPilot. Even some experienced trading veterans do not seem to get the point that frequencies do not matter. Jim Rogers, a “legendary” investor, made the following statement:

I don't buy options. Buying options is another way to go to the poorhouse. Someone did a study for the SEC and discovered that 90 percent of all options expire as losses. Well, I figured out that if 90 percent of all long option positions lost money, that meant that 90 percent of all short option positions make money. If I want to use options to be bearish, I sell calls.

Visibly, the statistic that 90% of all option positions lost money is meaningless, (i.e., the *frequency*) if we do not take into account *how much* money is made on average during the remaining 10%. If we make 50 times our bet on average when the option is in the money, then I can safely make the statement that buying options is another way to go to the palazzo rather than the poorhouse. Mr Jim Rogers seems to have gone very far in life for someone who does not distinguish between probability and expectation (strangely, he was the partner of George Soros, a complex man who thrived on rare events – more on him later).

One such rare event is the stock market crash of 1987, which made me as a trader and allowed me the luxury of becoming involved in all manner of scholarship. Nero of the smaller house in Chapter 1 aims to get out of harm's way by avoiding exposure to rare events – a mostly defensive approach. I am far more aggressive than Nero and go one step further; I have organized my career and business in such a way as to be able to benefit from them. In other words, I aim at profiting from the rare event, with my asymmetric bets.

SYMMETRY AND SCIENCE

In most disciplines, such asymmetry does not matter. Unfortunately, the techniques used in finance are often imported from other areas – finance

is still a young discipline (it is certainly not yet a “science”). People in most fields outside of finance do not have problems eliminating extreme values from their sample, when the difference in payoff between different outcomes is not significant, which is generally the case in education and medicine. A professor who computes the average of his students' grades removes the highest and lowest observations, which he would call *outliers*, and takes the average of the remaining ones, without this being an unsound practice. A casual weather forecaster does the same with extreme temperatures – an unusual occurrence might be deemed to skew the overall result (though we will see that this may turn out to be a mistake when it comes to forecasting future properties of the ice cap). So people in finance borrow the technique and ignore infrequent events, not noticing that the effect of a rare event can bankrupt a company.

Many scientists in the physical world are also subject to such foolishness, misreading statistics. One flagrant example is in the global-warming debate. Many scientists failed to notice it in its early stages as they removed from their sample the spikes in temperature, under the belief that these were not likely to recur. It may be a good idea to take out the extremes when computing the average temperatures for vacation scheduling. But it does not work when we study the physical properties of the weather. These scientists initially ignored the fact that these spikes, although rare, had the effect of adding disproportionately to the cumulative melting of the ice cap. Just as in finance, an event, although rare, that brings large consequences cannot just be ignored.

Figure 6.1 shows a series of points starting with an initial level W_0 and ending at the period concerned W_T . It can also be seen as the performance, hypothetical or realized, of your favorite trading strategy, the track record of an investment manager, the price of a foot of average Palazzo real estate in Renaissance Florence, the price series of the Mongolian stock market, or the difference between the U.S. and Mongolian stock markets. It is composed of a given number of sequential observations W_1, W_2 , etc., ordered in such a way that the one to the right comes *after* the one to the left.

If we were dealing with a deterministic world – that is, a world stripped of randomness (the right-column world in Table P.1 on page 3), and we knew with certainty that it was the case, things would be rather